DATASET-1 (9th May 2023 - 18th June 2023)

Predicted day - 18/6/2023 (Day 41st)

■ Hourly equatorial Dst values - source:- [link]

WDC for Geomagnetism, Kyoto
Hourly Equatorial Dst Values (PROVISIONAL)
JUNE 2023

unit=nT

DAY \ UT	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	-19	-15	-19	-21	-27	-22	-15	-21	-33	-30	-26	-19	-20	-27	-33	-43	-42	-45	-45	-36	-29	-27	-22	-20
2	-20	-25	-24	-18	-16	-17	-15	-10	-10	-13	-13	-13	-15	-17	-15	-17	-17	-16	-13	-11	-12	-15	-14	-10
3	-9	-8	-7	-4	-3	-2	-5	-8	-10	-8	-5	-5	-3	-2	-2	-3	-2	-2	-3	-3	-1	-1	0	2
4	5	13	19	21	16	12	10	6	2	4	3	1	-5	-2	2	1	2	6	4	-8	-15	-16	-13	-13
5	-14	-16	-13	-11	-11	-10	-8	-7	-6	-1	-3	-1	-1	-2	-1	-1	1	3	2	4	1	3	1	1
6	6	6	-1	-7	-4	-1	-2	-5	-4	-7	-7	-4	-3	-8	-9	-10	-8	-7	-8	-7	-5	-6	-7	-8
7	-6	-4	-4	-1	-1	-3	-1	2	1	-4	-4	-4	-6	-3	-1	-2	-2	0	1	1	0	-2	-2	0
8	-1	-2	1	5	4	4	1	-2	-5	-8	-8	-9	-9	-8	-5	-7	-5	-2	-2	-1	1	4	6	4
9	3	1	-1	-4	-5	-6	-5	-4	-1	0	2	2	3	4	6	9	8	5	4	3	3	4	4	1
10	-2	-2	2	4	6	6	7	7	8	8	7	7	7	5	2	1	0	0	2	6	11	14	15	15
11	11	14	14	15	13	10	12	18	22	24	22	18	15	15	13	11	9	1	0	1	4	3	8	10
12	6	5	2	2	1	0	1	4	2	2	5	4	2	1	-5	-5	-5	-5	-5	-4	1	2	1	-3
13	-4	-3	-1	1	0	1	2	1	3	4	5	3	4	3	3	1	-1	-7	-6	-4	-1	-2	-1	-1
14	-4	-9	-11	-6	-4	-3	-2	0	1	3	3	3	1	-2	-1	-2	-2	-4	-5	-5	-3	-2	-1	0
15	1	0	2	1	-3	-3	-4	-6	1	4	7	4	4	20	19	19	-1	-11	-29	-35	-41	-37	-35	-41
16	-41	-45	-45	-54	-53	-61	-65	-59	-68	-72	-57	-52	-48	-51	-52	-59	-60	-58	-58	-57	-56	-49	-44	-42
17	-36	-35	-36	-34	-34	-38	-38	-35	-35	-34	-31	-28	-28	-29	-28	-33	-33	-32	-31	-32	-33	-33	-33	-35
18	-33	-32	-27	-30	-31	-34	-33	-28	-28	-28	-28	-30	-29	-29	-30	-33	-41	-47	-50	-47	-44	-41	-38	-37
19	-36	-37	-32	-29	-23	-25	-26	-23	-19	-15	-15	-15	-14	-17	-19	-22	-24	-28	-32	-34	-32	-25	-29	-33
20	-33	-28	-25	-21	-20	-21	-23	-20	-18	-18	-16	-17	-20	-22	-24	-25	-26	-25	-24	-21	-18	-18	-21	-21

18h June 2023 - Minimum Dst Value: **-50 nT (at UT 19)**

■ Geomagnetic indices (kp, ap) , source:- [link]

column 8 (Kp) and column 9 (ap) for all 8 three-hour intervals of that UTC day

18th June 2023

2023	06	18	00.0	01.50	33406.00000	33406.06250	2.667	12	1
2023	06	18	03.0	04.50	33406.12500	33406.18750	2.333	9	1
2023	06	18	06.0	07.50	33406.25000	33406.31250	1.333	5	1
2023	06	18	09.0	10.50	33406.37500	33406.43750	1.667	6	1
2023	06	18	12.0	13.50	33406.50000	33406.56250	1.667	6	1
2023	06	18	15.0	16.50	33406.62500	33406.68750	2.333	9	1
2023	06	18	18.0	19.50	33406.75000	33406.81250	3.000	15	1
2023	06	18	21.0	22.50	33406.87500	33406.93750	2.000	7	1

Peak Kp : 3.000 (3o) , Peak ap : 15

■ Conclusion -

June 18, 2023, was a quiet day. The peak Kp here reached only **3o**, which represents completely unsettled to minor active baseline conditions. The Dst index dipped to a minimum of -50 nT, indicating a very minor, borderline negligible magnetospheric disturbance.

DATASET-2 (31st March 2024 - 10th May 2024)

Predicted day 1- 9/5/2024 (Day 40th)

Predicted day 2- 10/5/204 (Day 41st)

■ Hourly equatorial Dst values - source:- link

WDC for Geomagnetism, Kyoto
Hourly Equatorial Dst Values (PROVISIONAL)
MAY 2024

unit=nT

DAY \ UT	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	-19	-26	-21	-7	-5	-3	-5	-4	-2	0	-3	-4	-6	-5	-4	1	3	6	1	5	12	11	13	10
2	7	5	12	13	13	13	8	6	10	16	13	3	-3	-11	-11	-43	-71	-70	-82	-89	-88	-67	-44	-53
3	-55	-53	-47	-54	-51	-49	-47	-44	-38	-36	-33	-32	-31	-29	-30	-27	-27	-29	-30	-32	-28	-22	-17	-17
4	-18	-18	-17	-13	-10	-8	-7	-8	-9	-10	-10	-9	-9	-11	-11	-12	-12	-10	-9	-8	-9	-8	-3	2
5	4	2	3	0	-5	-7	-5	2	7	6	8	13	18	19	22	22	18	15	14	10	-1	-18	-33	-39
6	-33	-43	-35	-34	-33	-27	-17	-17	-9	-6	-3	-4	-1	1	-3	1	-4	-9	-12	-13	-13	-10	-8	-5
7	-6	-6	-4	-7	-7	-3	2	6	4	1	-1	-4	-6	-7	-8	-8	-10	-15	-12	-8	-2	1	5	4
8	3	5	7	5	6	9	9	9	8	9	7	7	6	4	6	5	5	3	3	5	7	11	15	14
9	15	17	18	15	15	14	14	13	14	15	15	17	18	21	21	18	18	20	22	22	21	20	20	17
10	12	10	12	21	16	7	9	16	13	15	13	3	-1	1	10	23	25	66	-33	-131	-157	-277	-339	-308

9h May 2024 - Minimum Dst Value: **+13 nT (at UT 8)** , Maximum Dst Value: **+22 nT (at UT 19 & 20)**

10th May 2024 - Minimum Dst Value: **339 nT (at UT 23)** , Maximum Dst Value: **+66 nT (at UT 18)**

■ Geomagnetic indices (kp, ap) , source:- link

column 8 (Kp) and column 9 (ap) for all 8 three-hour intervals of that UTC day

9th May 2024

```
2024 05 09 00.0 01.50 33732.00000 33732.06250  0.667    3 1
2024 05 09 03.0 04.50 33732.12500 33732.18750  1.333    5 1
2024 05 09 06.0 07.50 33732.25000 33732.31250  0.667    3 1
2024 05 09 09.0 10.50 33732.37500 33732.43750  1.333    5 1
2024 05 09 12.0 13.50 33732.50000 33732.56250  1.000    4 1
2024 05 09 15.0 16.50 33732.62500 33732.68750  1.000    4 1
2024 05 09 18.0 19.50 33732.75000 33732.81250  2.333    9 1
2024 05 09 21.0 22.50 33732.87500 33732.93750  2.000    7 1
```

Peak Kp : 2.333 (2+) , Peak ap : 9

10th May 2024

```
2024 05 10 00.0 01.50 33733.00000 33733.06250  2.667   12 1
2024 05 10 03.0 04.50 33733.12500 33733.18750  2.667   12 1
2024 05 10 06.0 07.50 33733.25000 33733.31250  2.333    9 1
2024 05 10 09.0 10.50 33733.37500 33733.43750  2.000    7 1
2024 05 10 12.0 13.50 33733.50000 33733.56250  3.667   22 1
2024 05 10 15.0 16.50 33733.62500 33733.68750  7.667  179 1
2024 05 10 18.0 19.50 33733.75000 33733.81250  8.667  300 1
2024 05 10 21.0 22.50 33733.87500 33733.93750  8.667  300 1
```

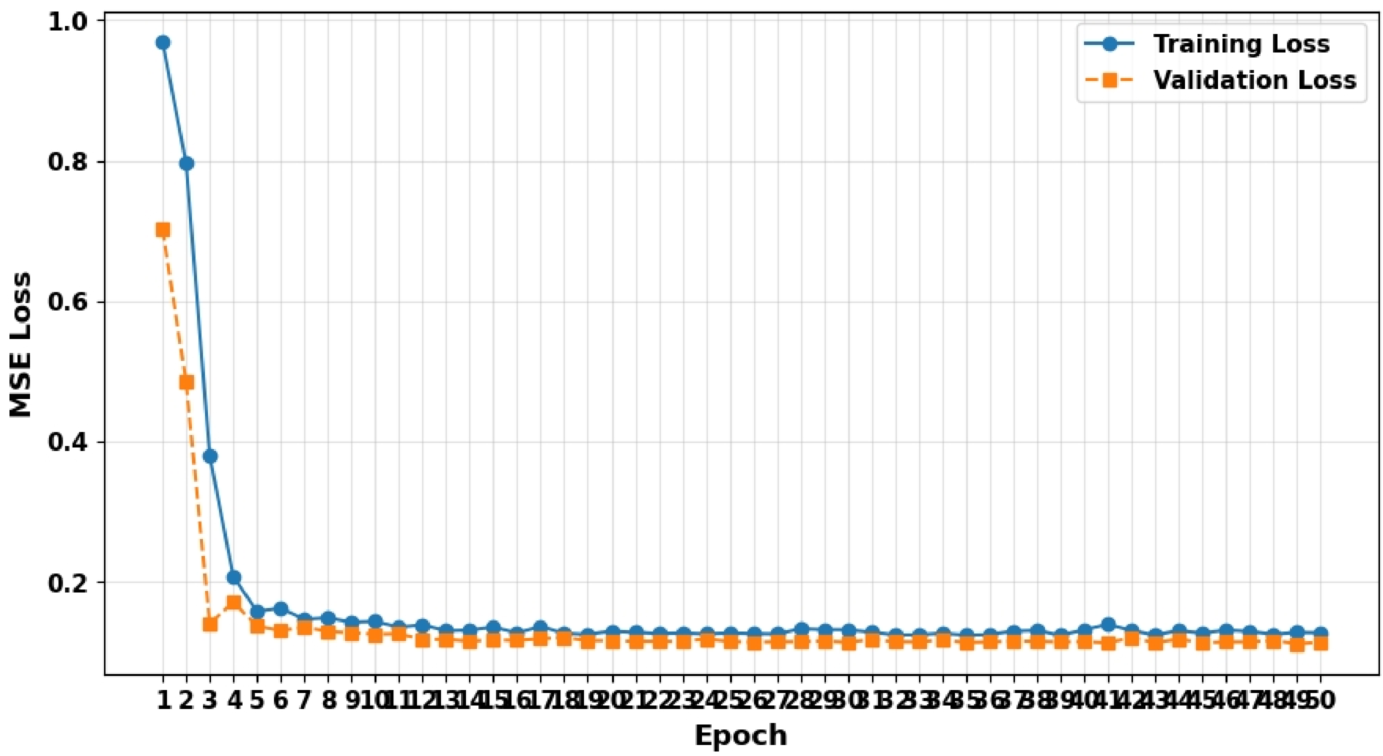
Peak Kp : 8.667(9-) , Peak ap : 300

■ Conclusion -

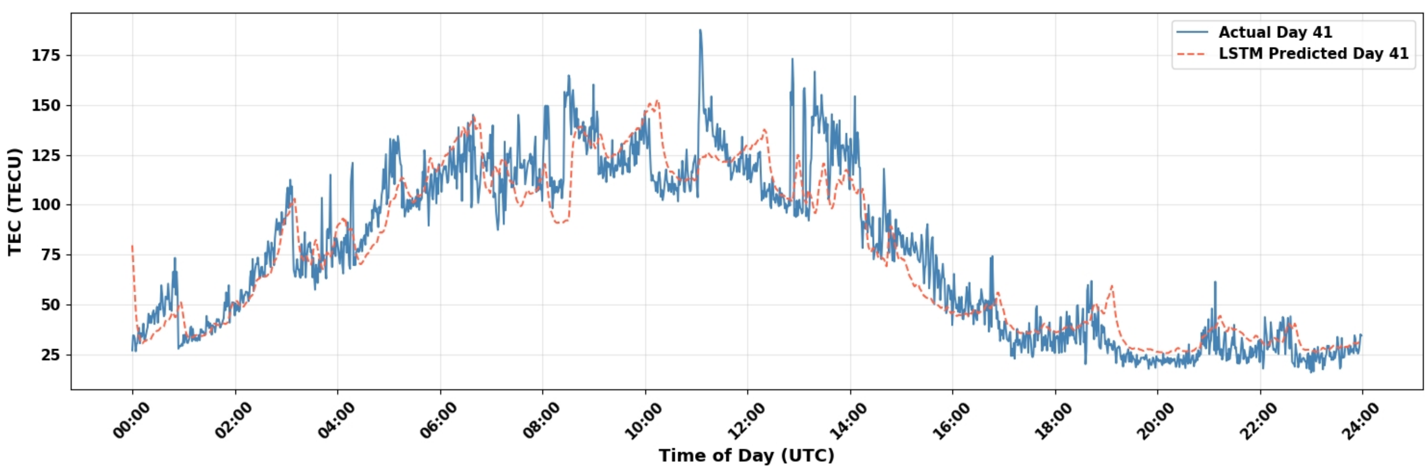
May 9, 2024, was completely quiet. The maximum Kp reached was only **2+** (well below the G3 threshold of 7).The Dst values remained positive (+13 nT to +22 nT), indicating a completely normal, stable magnetosphere.

May 10, 2024, shattered the G3 and G4 thresholds completely. It was a **G5 (Extreme) Geomagnetic Storm day**

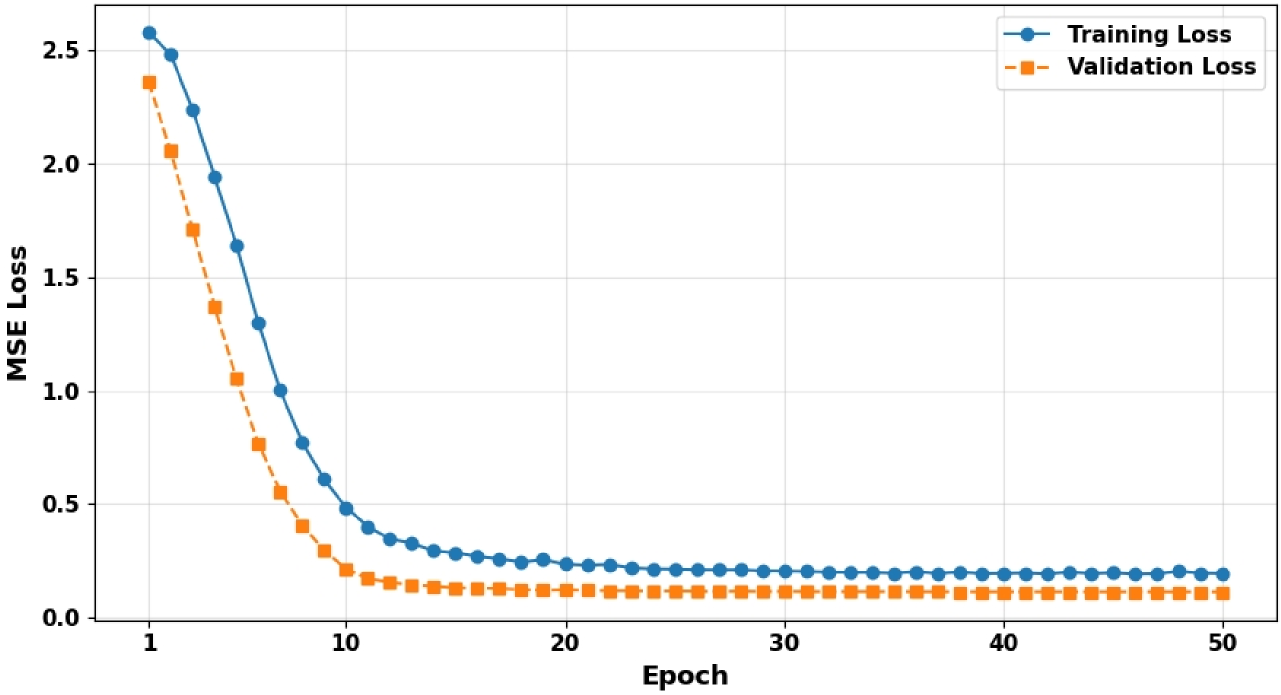
Plots - DataSet 1 - 18th June 2023



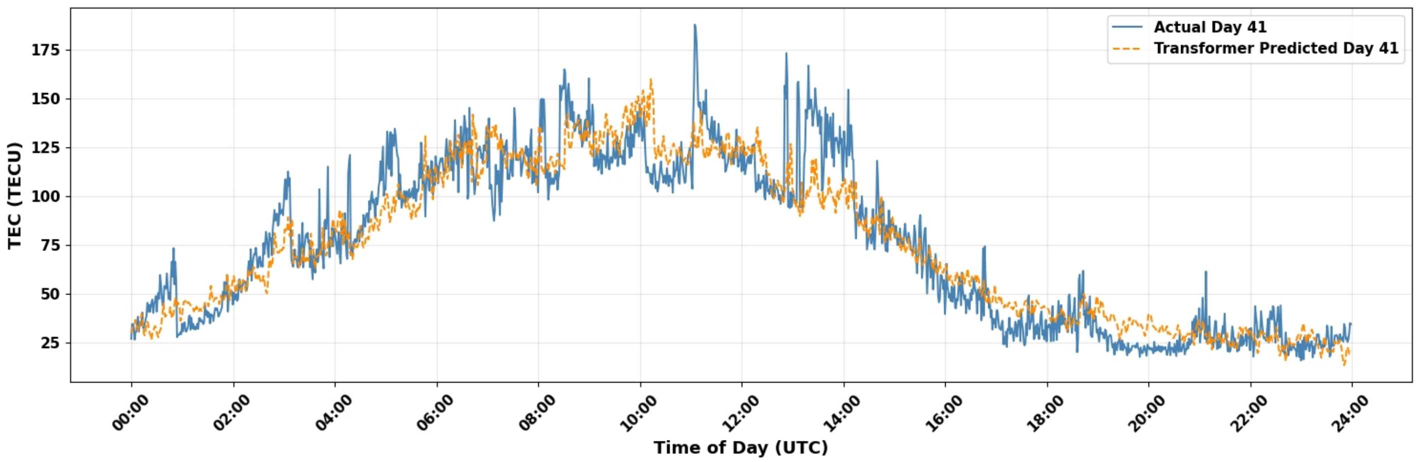
Plot 1- (LSTM) Loss vs Validation



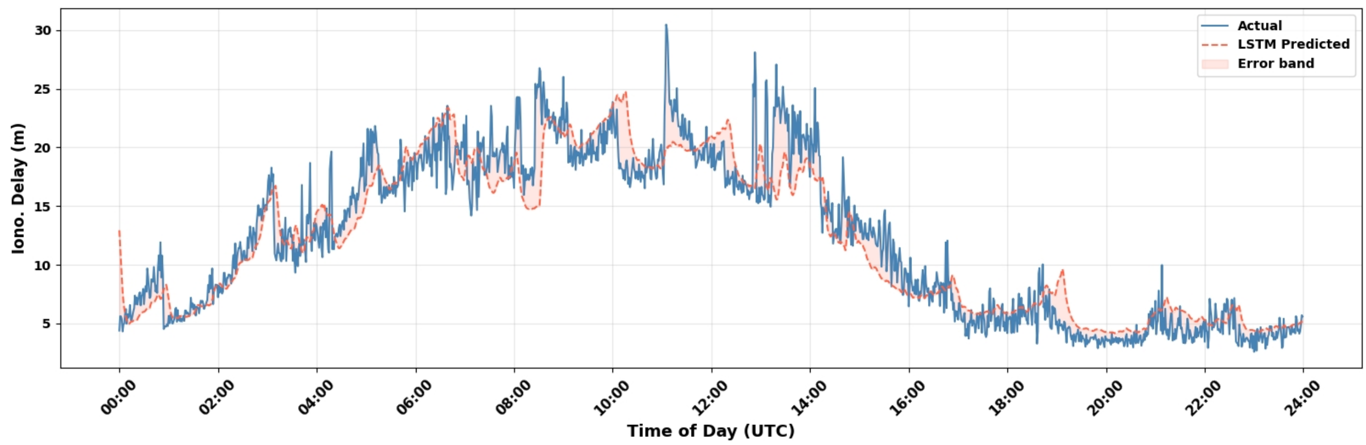
Plot 2- (LSTM) Prediction - 18th June 2023



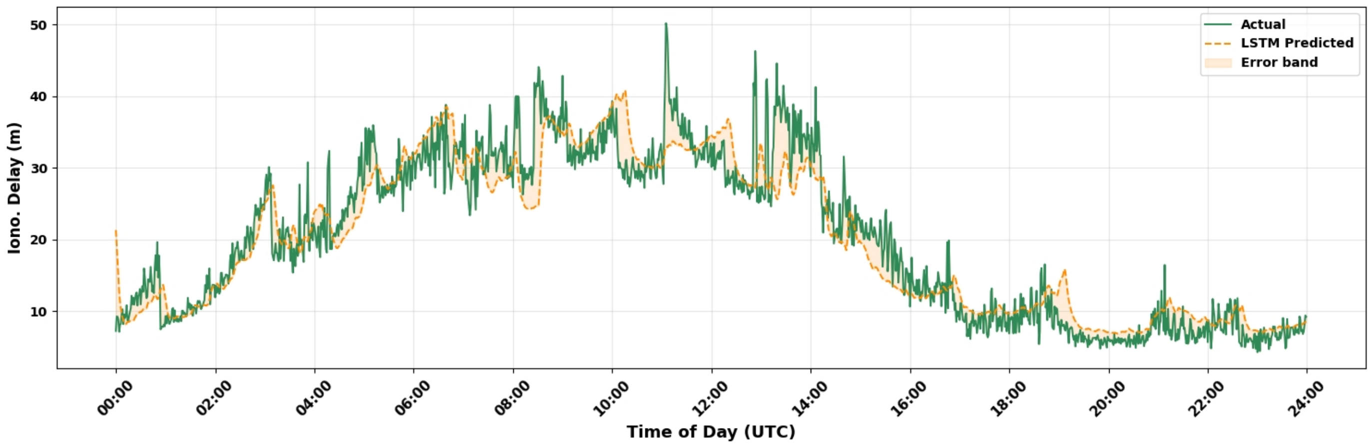
Plot 3- (Transformer) Loss vs Validation - 18th June 2023



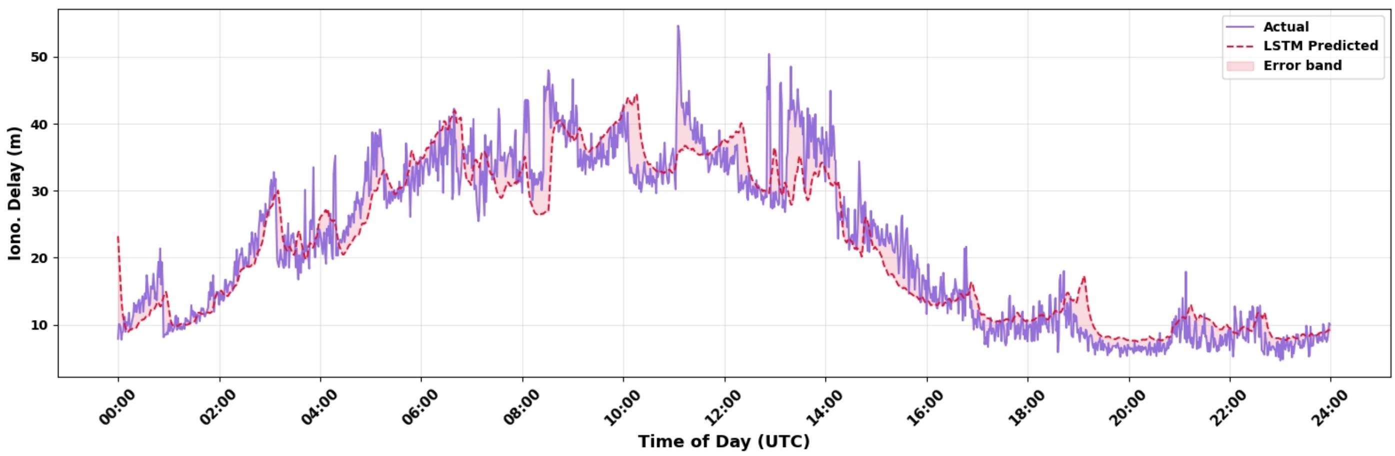
Plot 4- (Transformer) Prediction - 18th June 2023



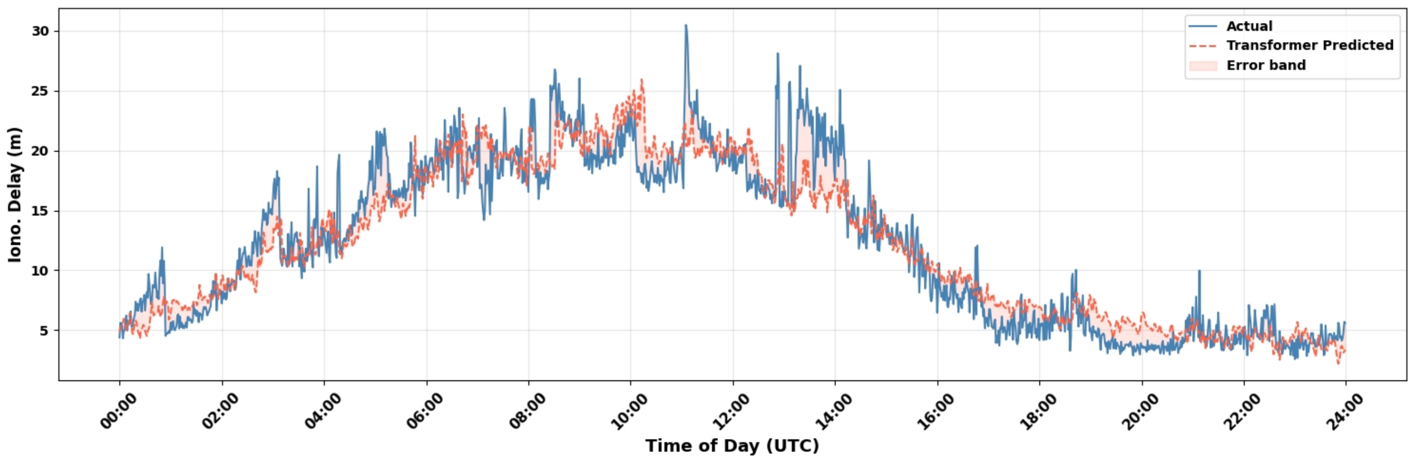
Plot 5- Ionospheric delay (LSTM) 18th June 2023 for L1 Frequency



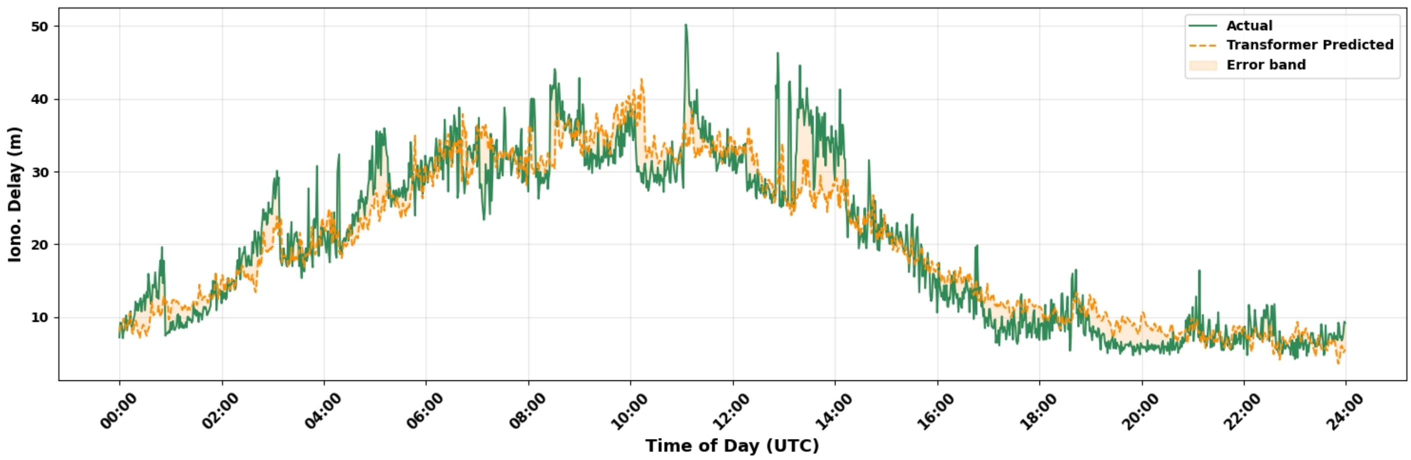
Plot 6- Ionospheric delay (LSTM) 18th June 2023 for L2 Frequency



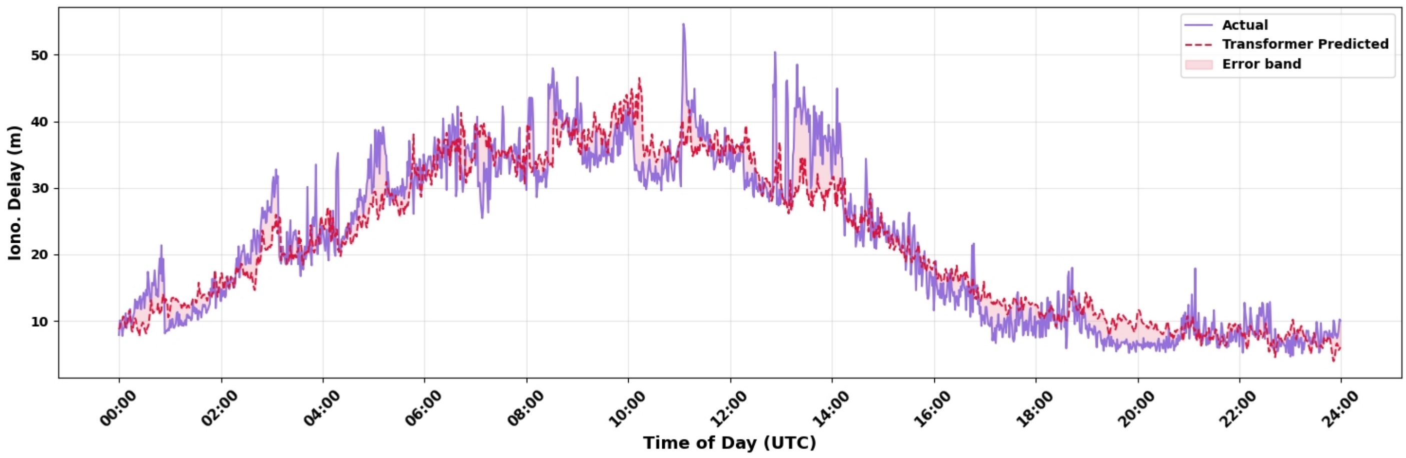
Plot 7- Ionospheric delay (LSTM) 18th June 2023 for L5 Frequency



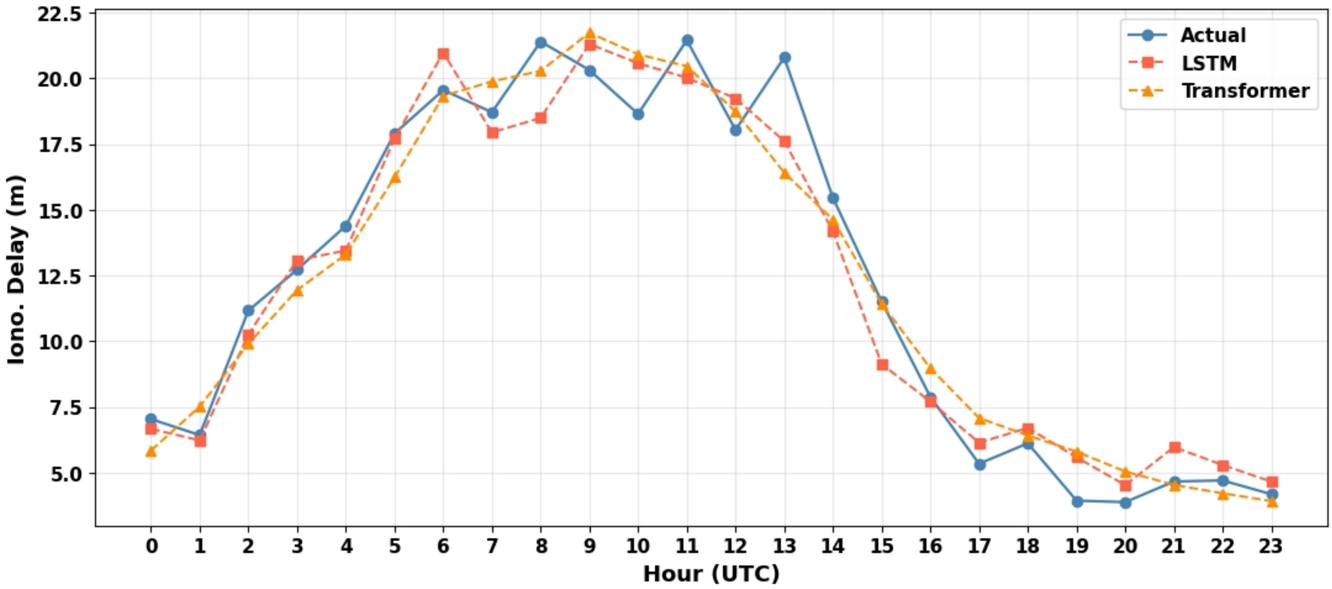
Plot 8- Ionospheric delay (Transformer) 18th June 2023 for L1 Frequency



Plot 9- Ionospheric delay (Transformer) 18th June 2023 for L2 Frequency

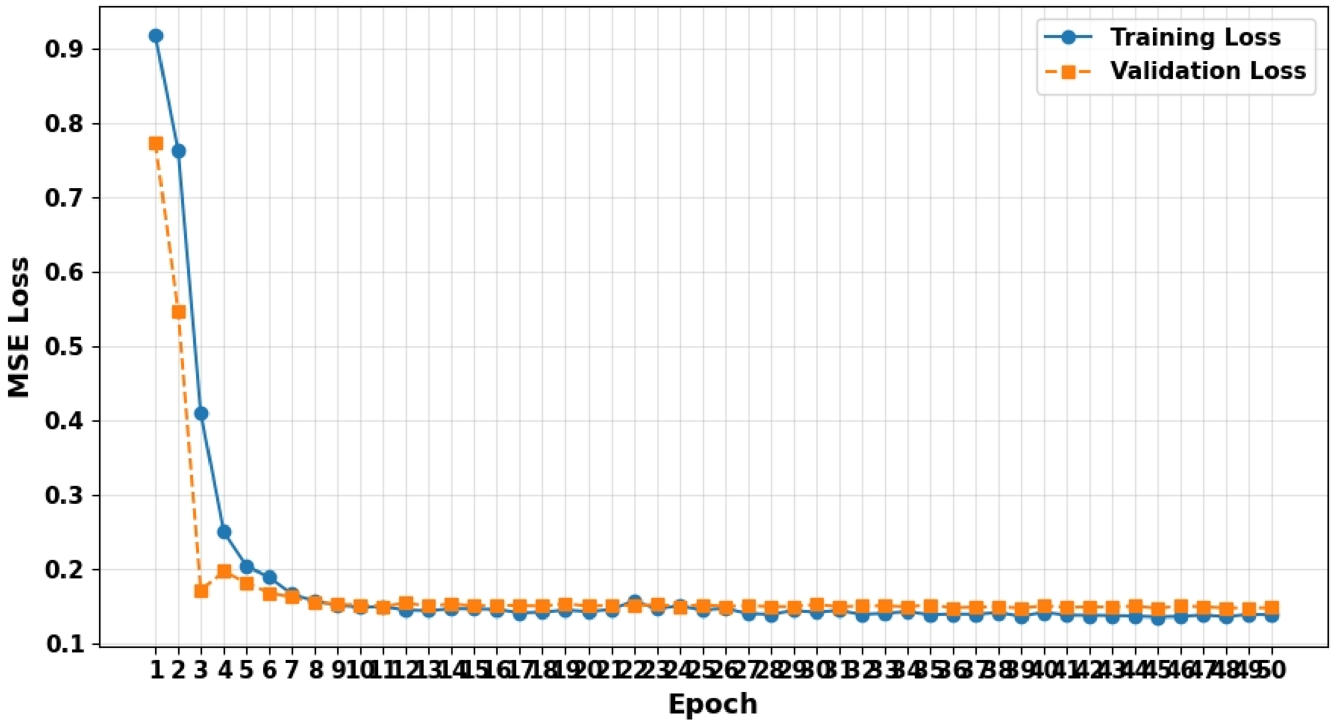


Plot 10- Ionospheric delay (Transformer) 18th June 2023 for L5 Frequency

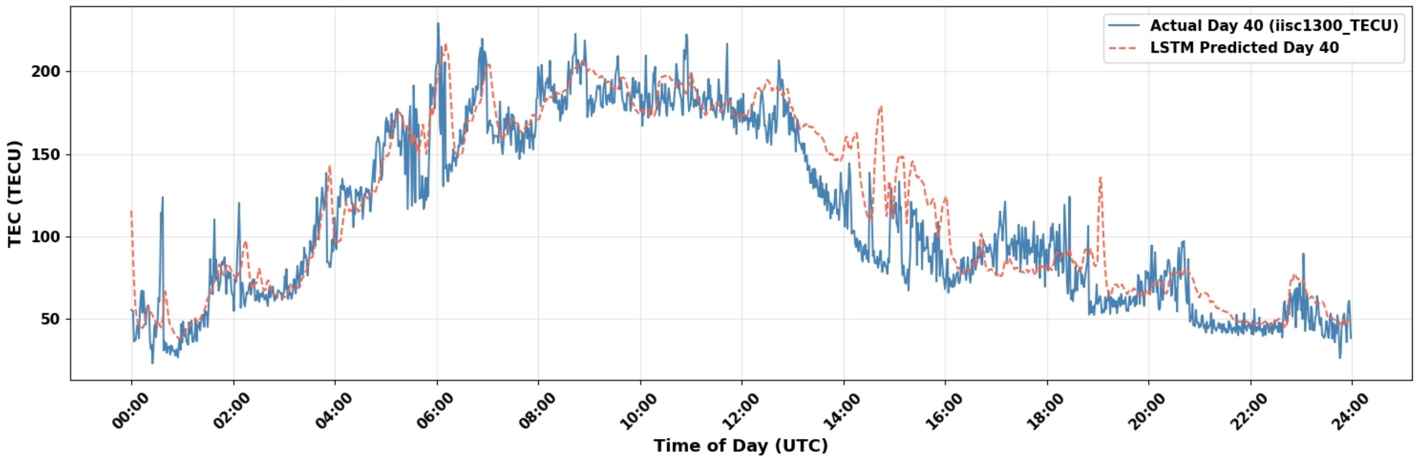


Plot 11 - Diurnal hourly profile — 18th June 2023

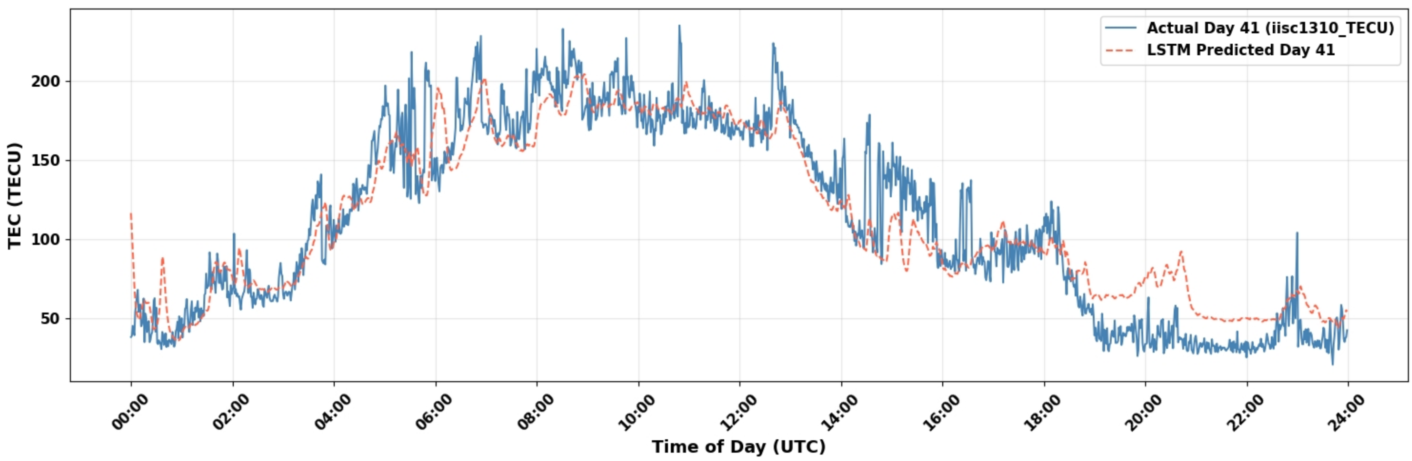
Plots - DataSet 2 - 9th May 2024 & 10th May 2024



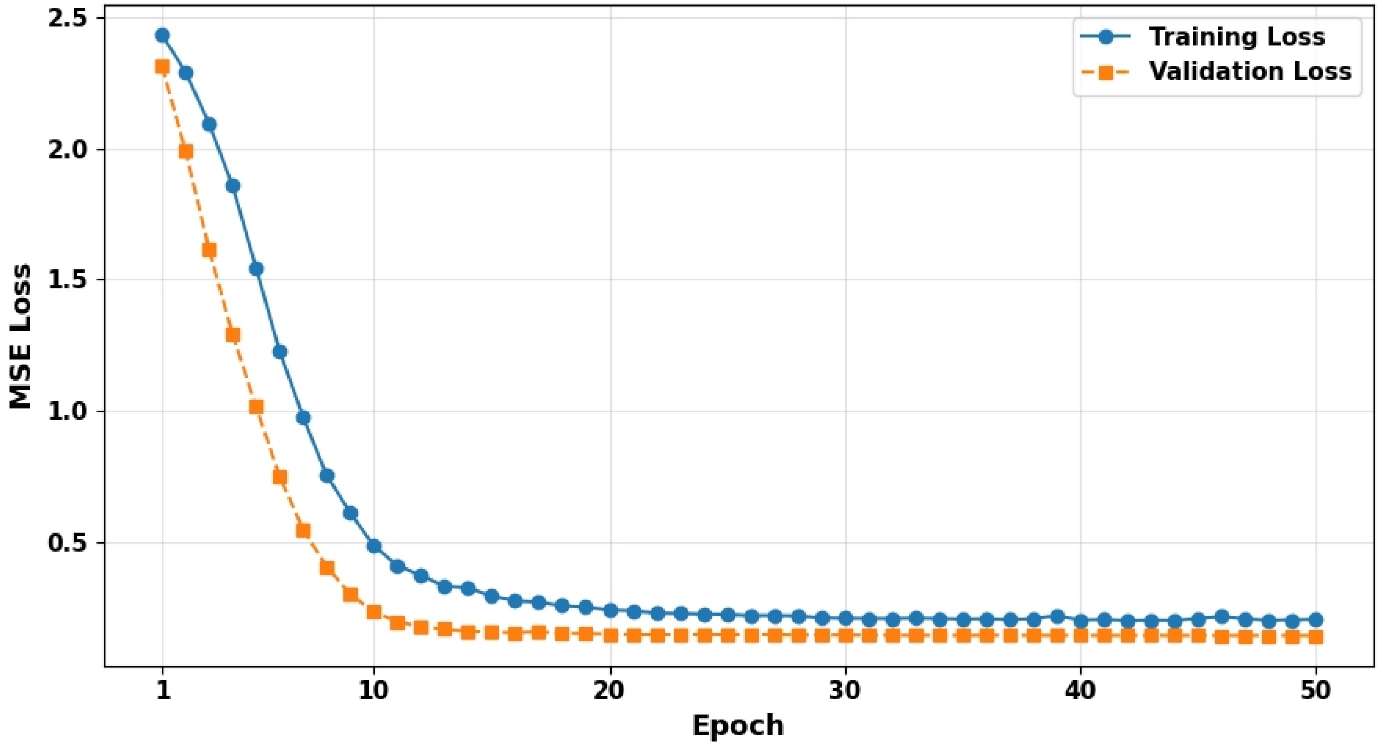
Plot 12- (LSTM) Loss vs Validation



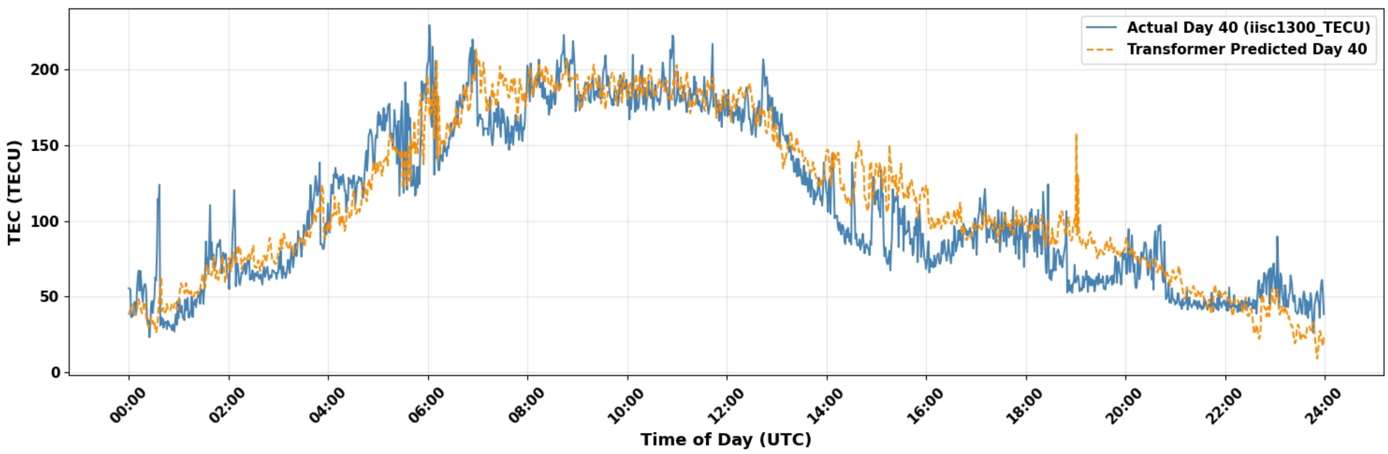
Plot 13- (LSTM) Prediction - 9th May 2024



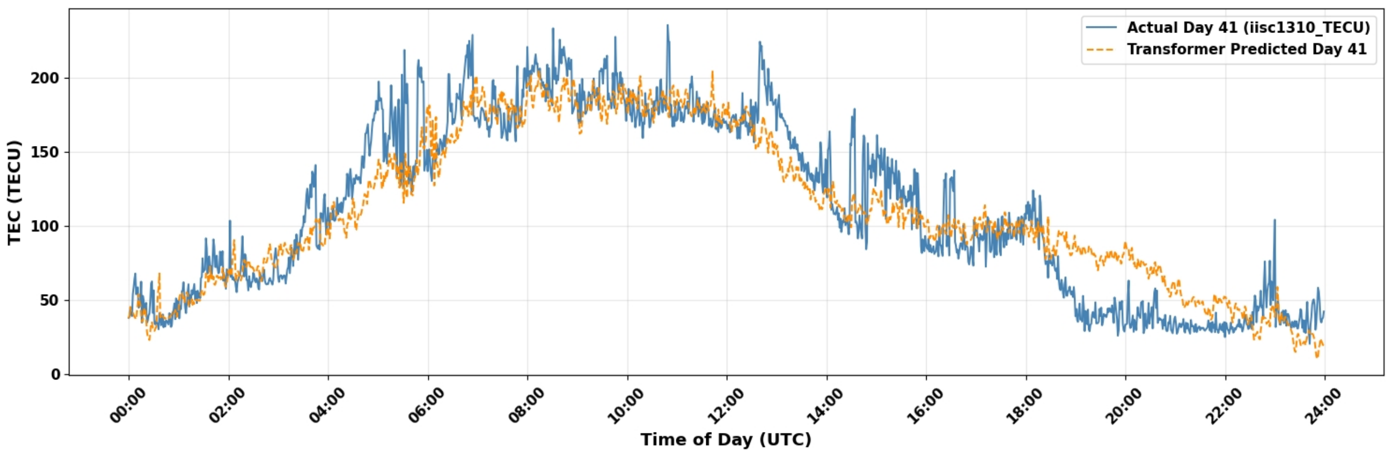
Plot 14- (LSTM) Prediction -10th May 2024



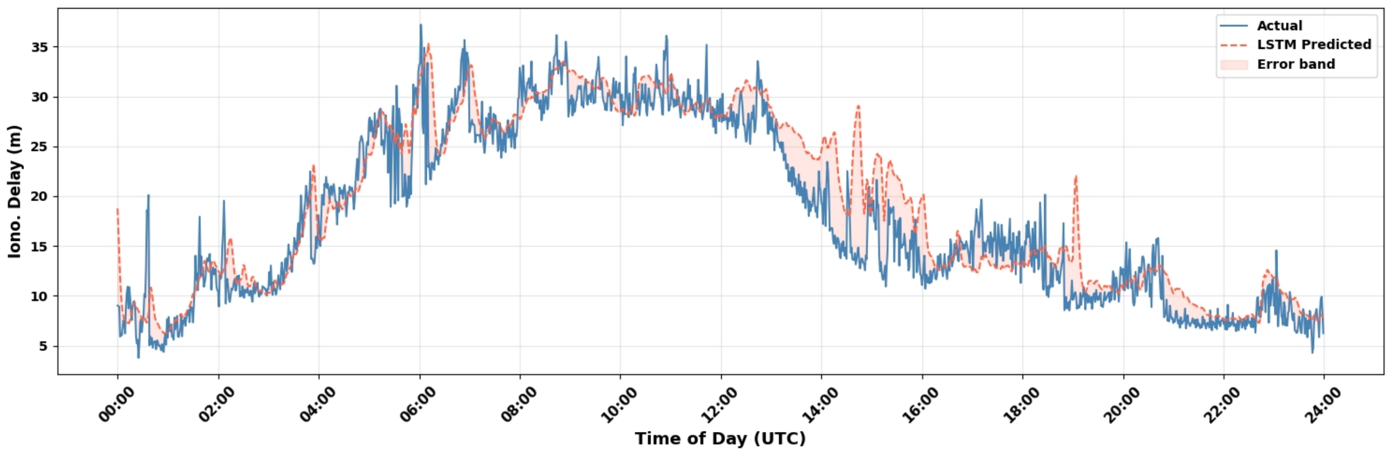
Plot 15- (Transformer) Loss vs Validation



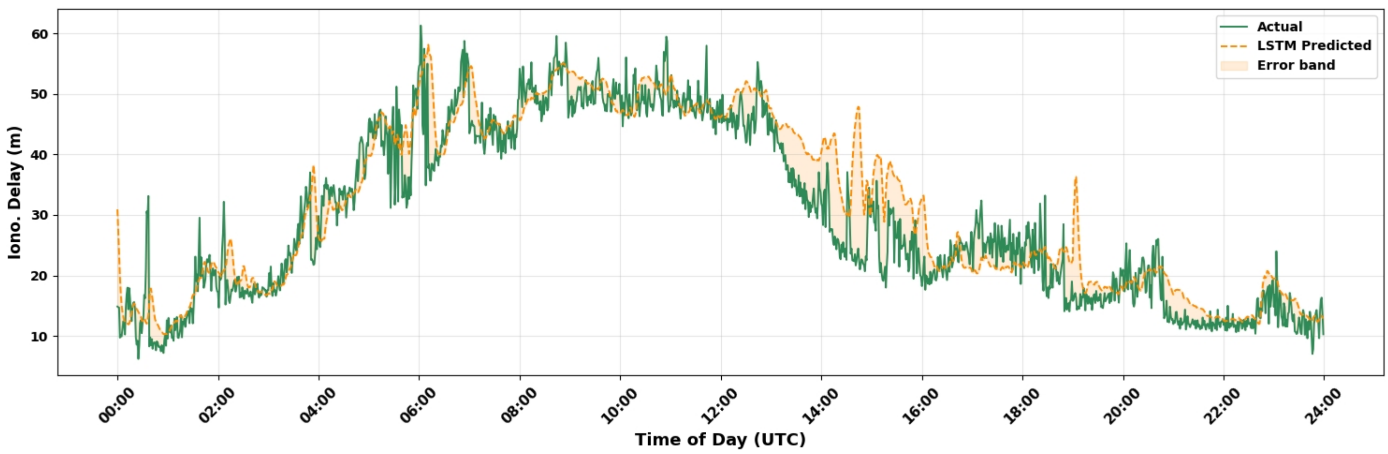
Plot 16- (Transformer) Prediction - 9th May 2024



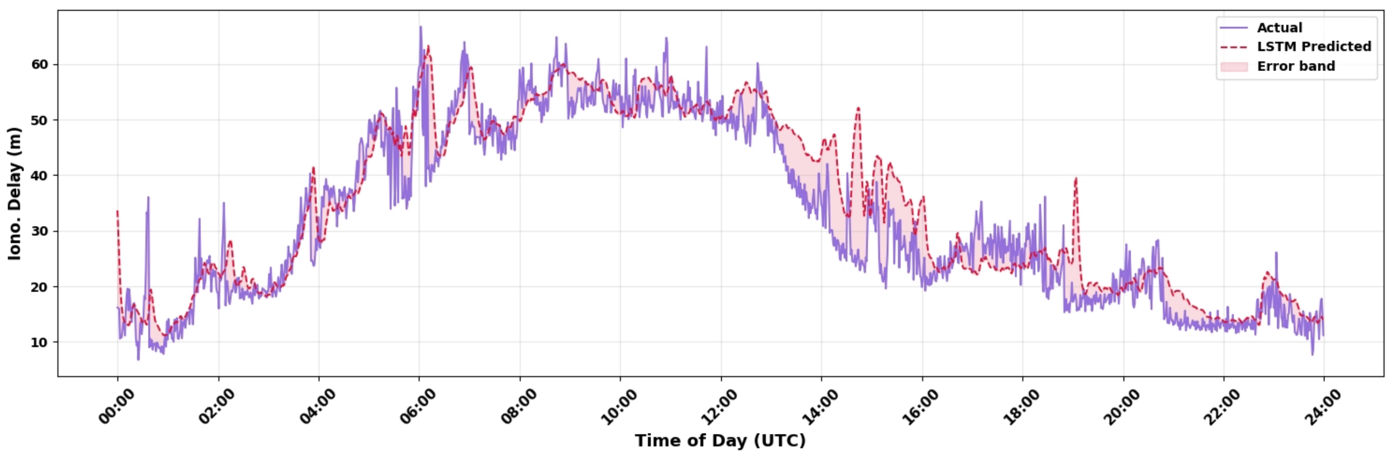
Plot 17- (Transformer) Prediction -10th May 2024



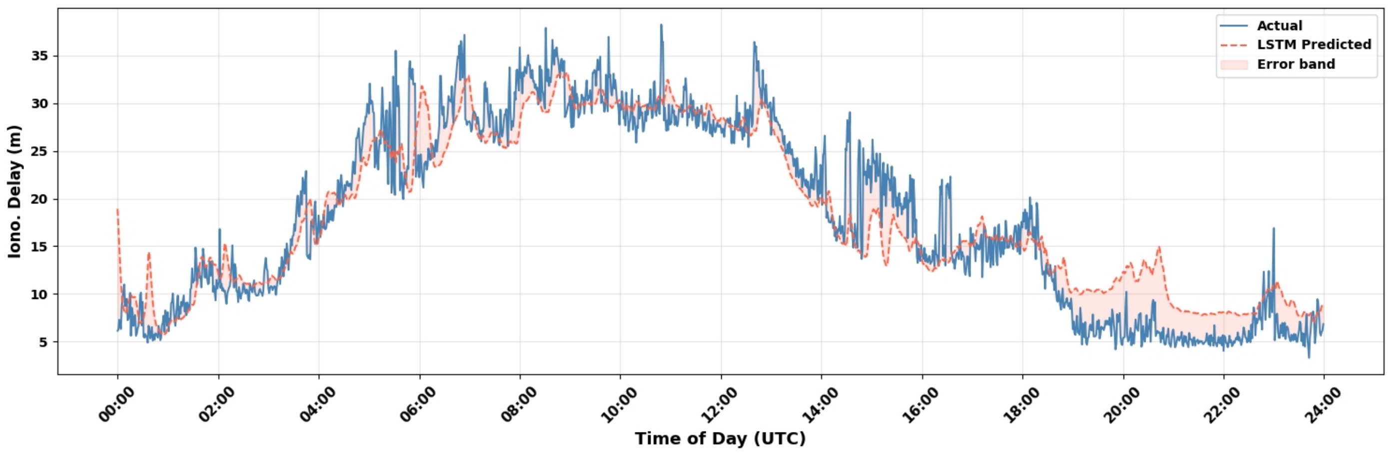
Plot 18- Ionospheric delay (LSTM) 9h May 2024 for L1 Frequency



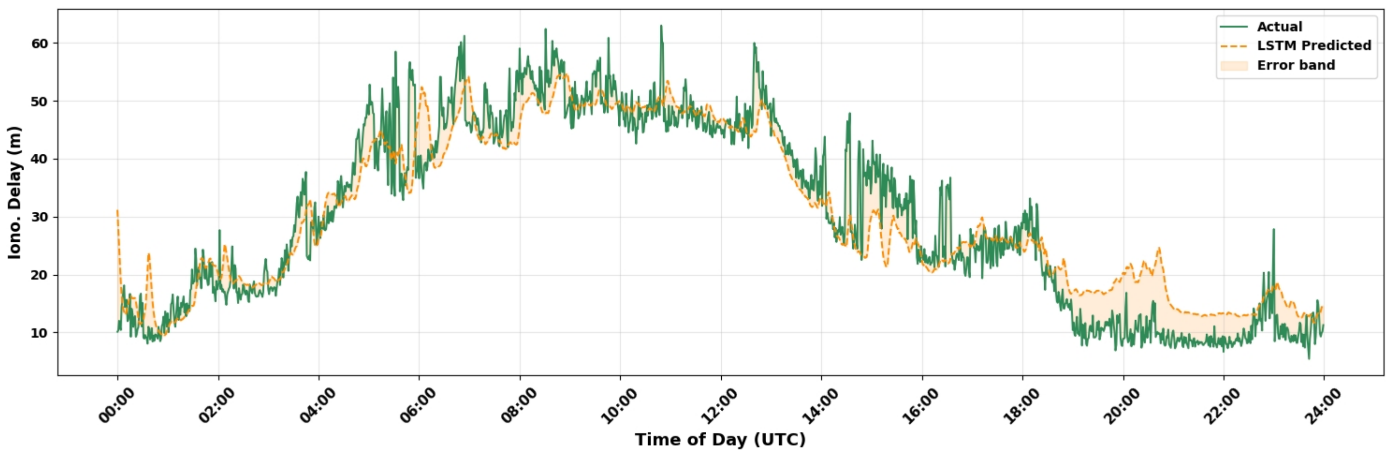
Plot 19- Ionospheric delay (LSTM) 9h May 2024 for L2 Frequency



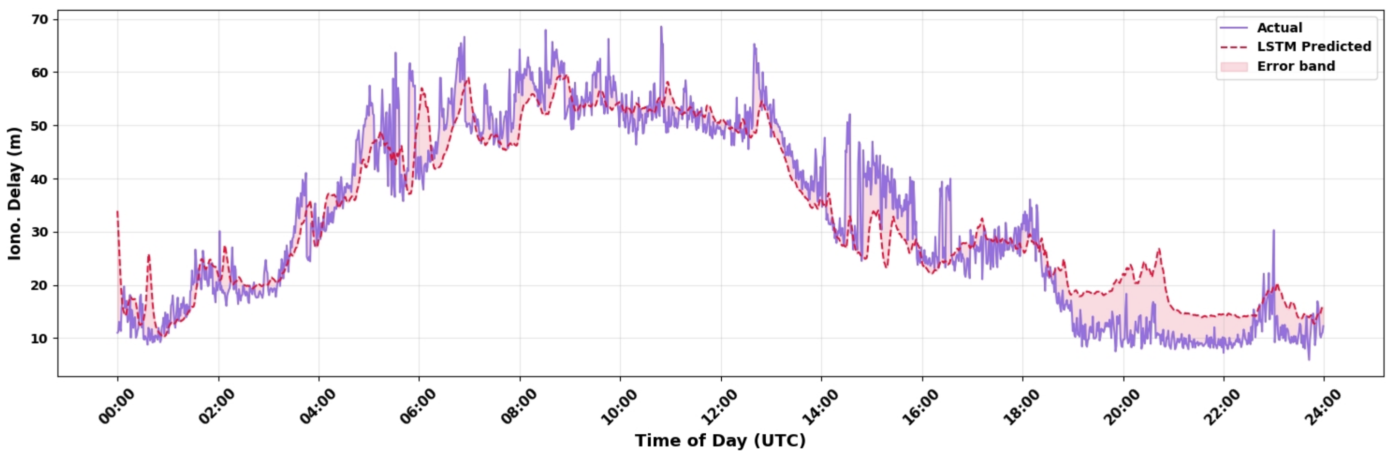
Plot 20- Ionospheric delay (LSTM) 9h May 2024 for L5 Frequency



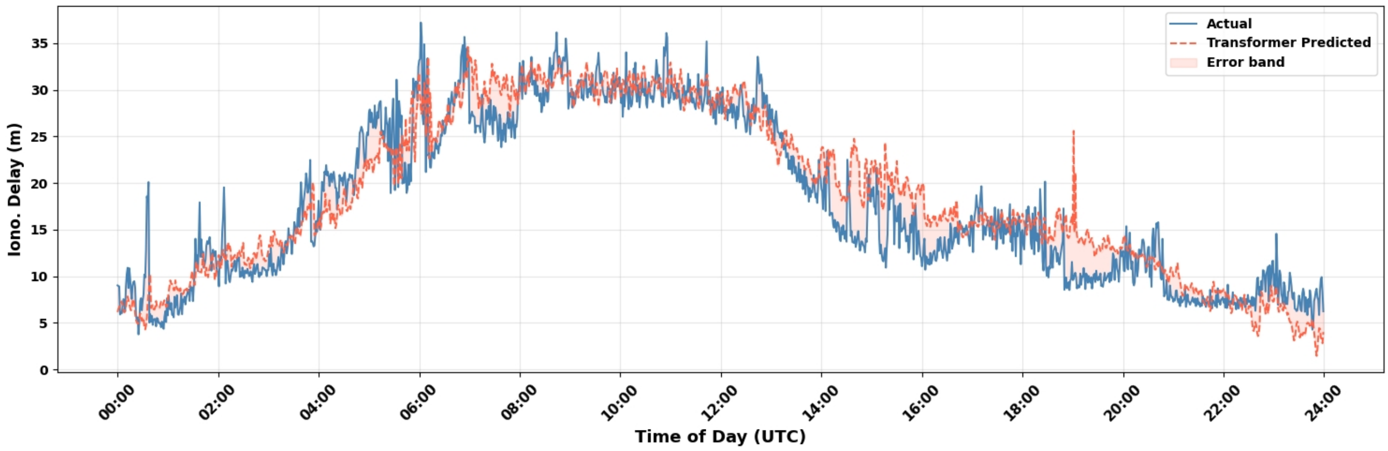
Plot 21- Ionospheric delay (LSTM) 10h May 2024 for L1 Frequency



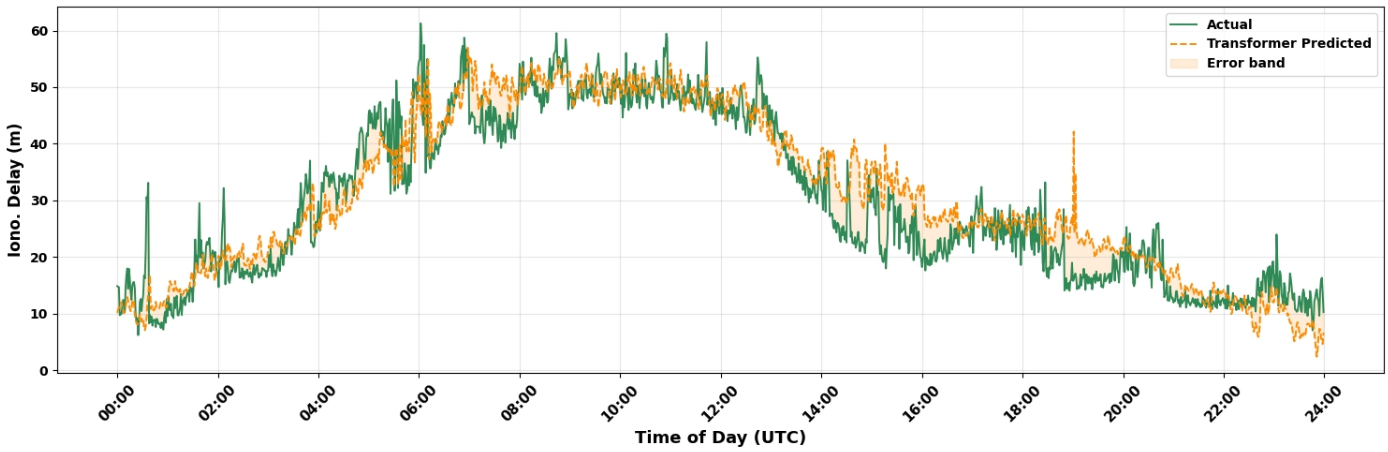
Plot 22- Ionospheric delay (LSTM) 10h May 2024 for L2 Frequency



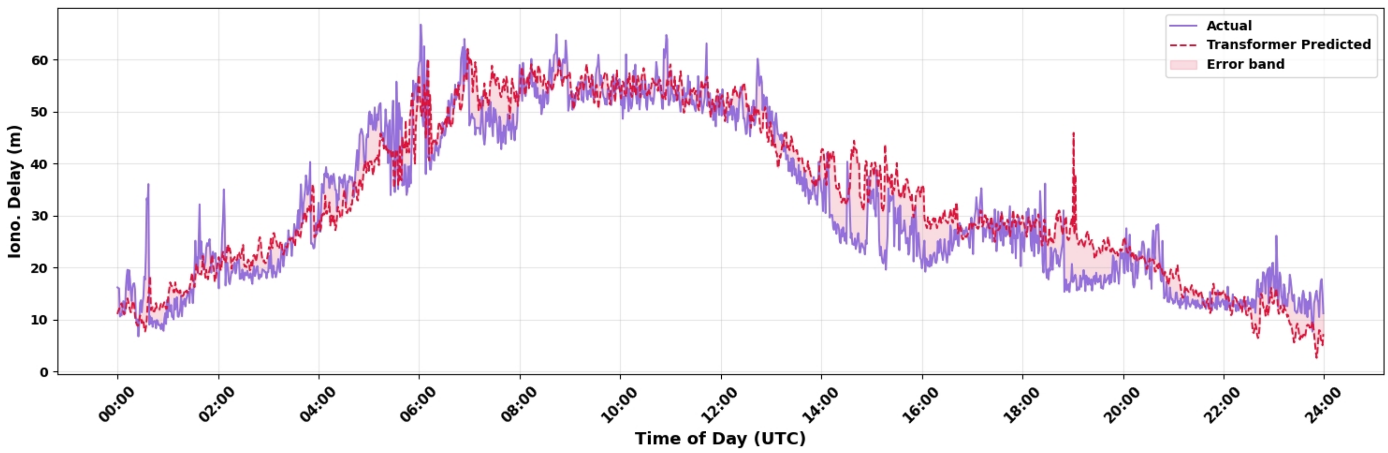
Plot 23- Ionospheric delay (LSTM) 10h May 2024 for L5 Frequency



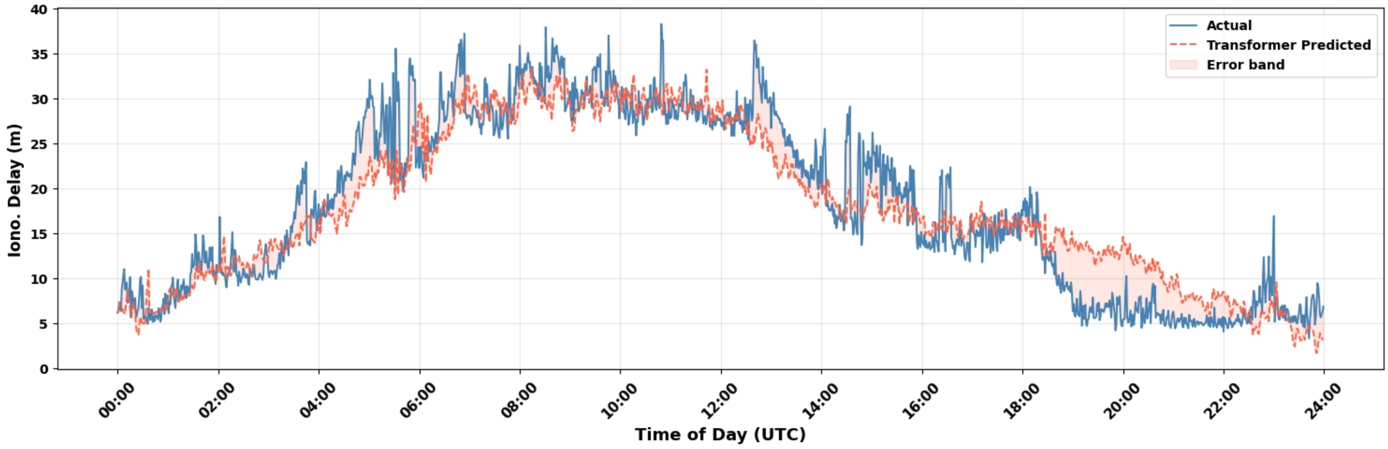
Plot 24- Ionospheric delay (Transformer) 9h May 2024 for L1 Frequency



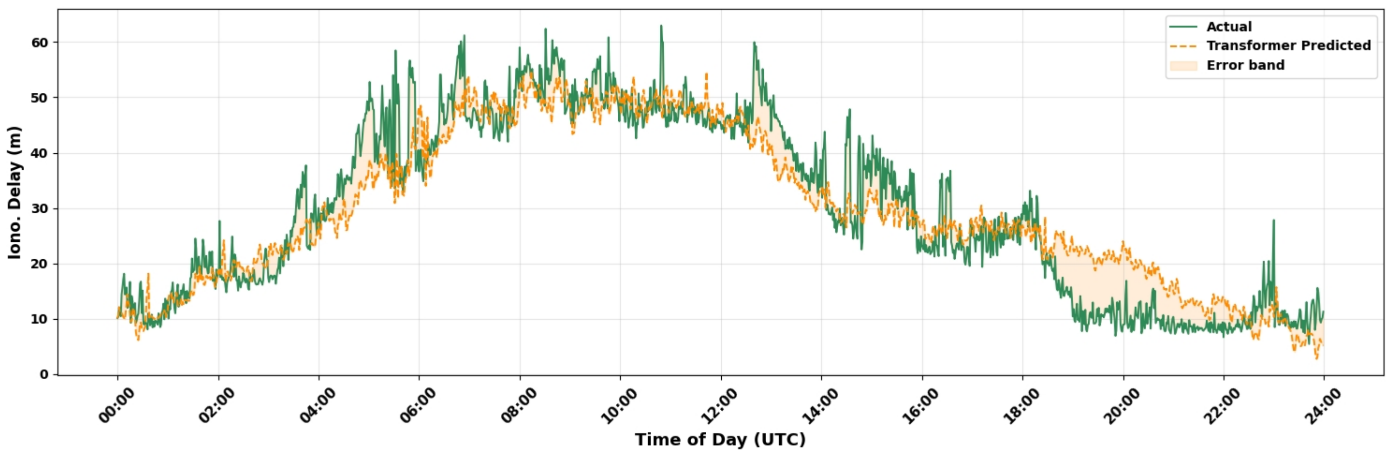
Plot 25- Ionospheric delay (Transformer) 9h May 2024 for L2 Frequency



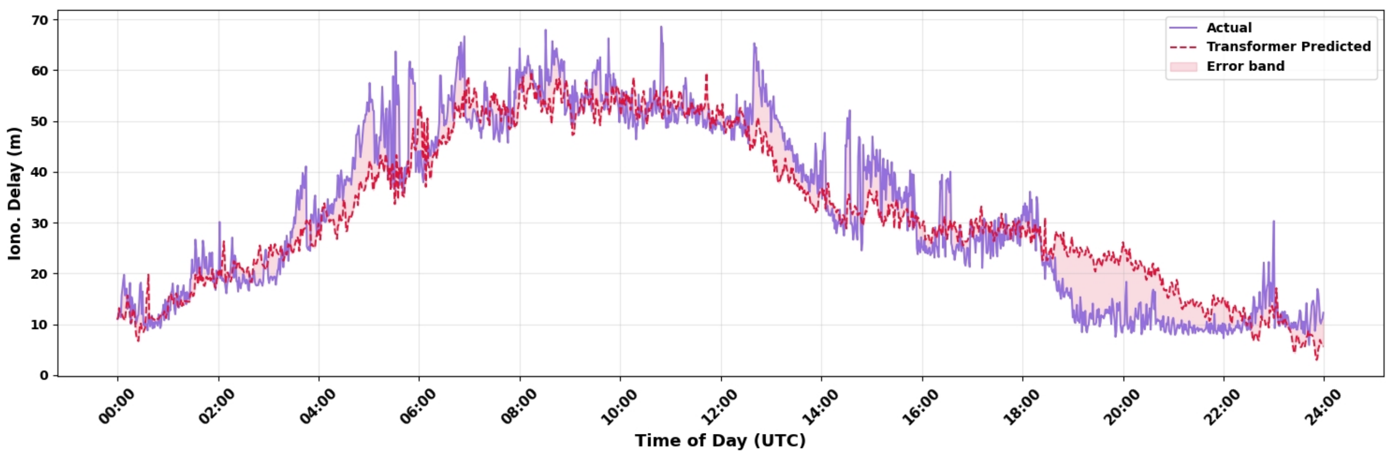
Plot 26- Ionospheric delay (Transformer) 9h May 2024 for L5 Frequency



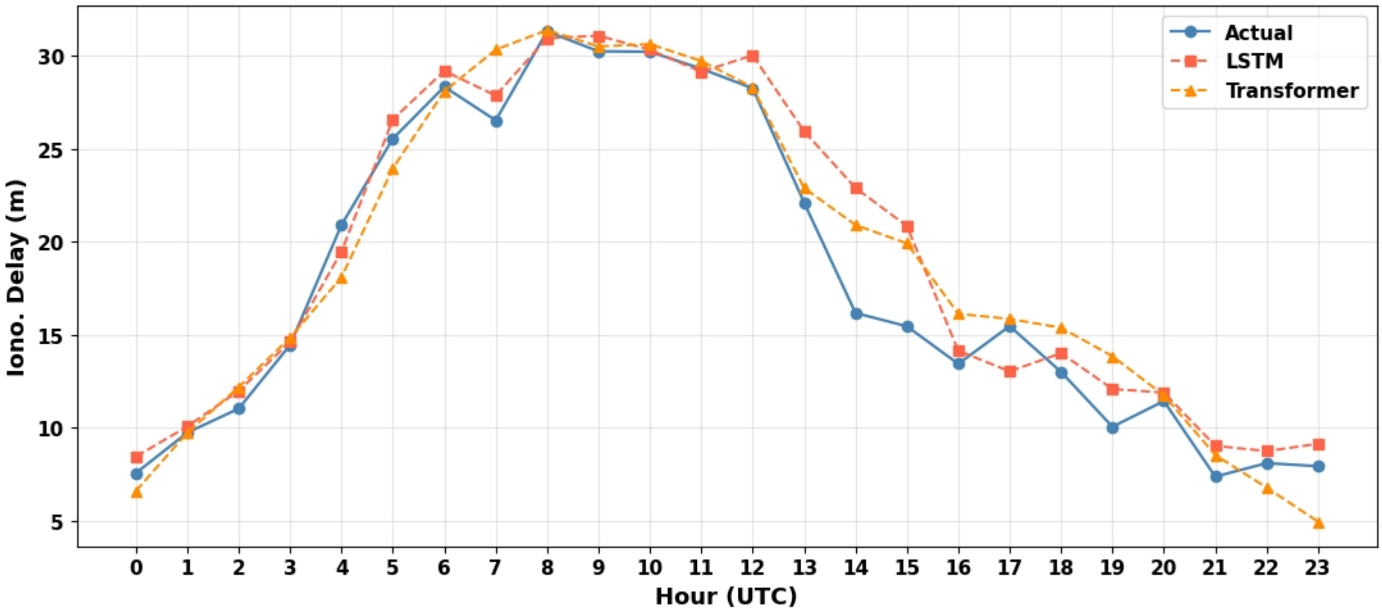
Plot 27- Ionospheric delay (Transformer) 10h May 2024 for L1 Frequency



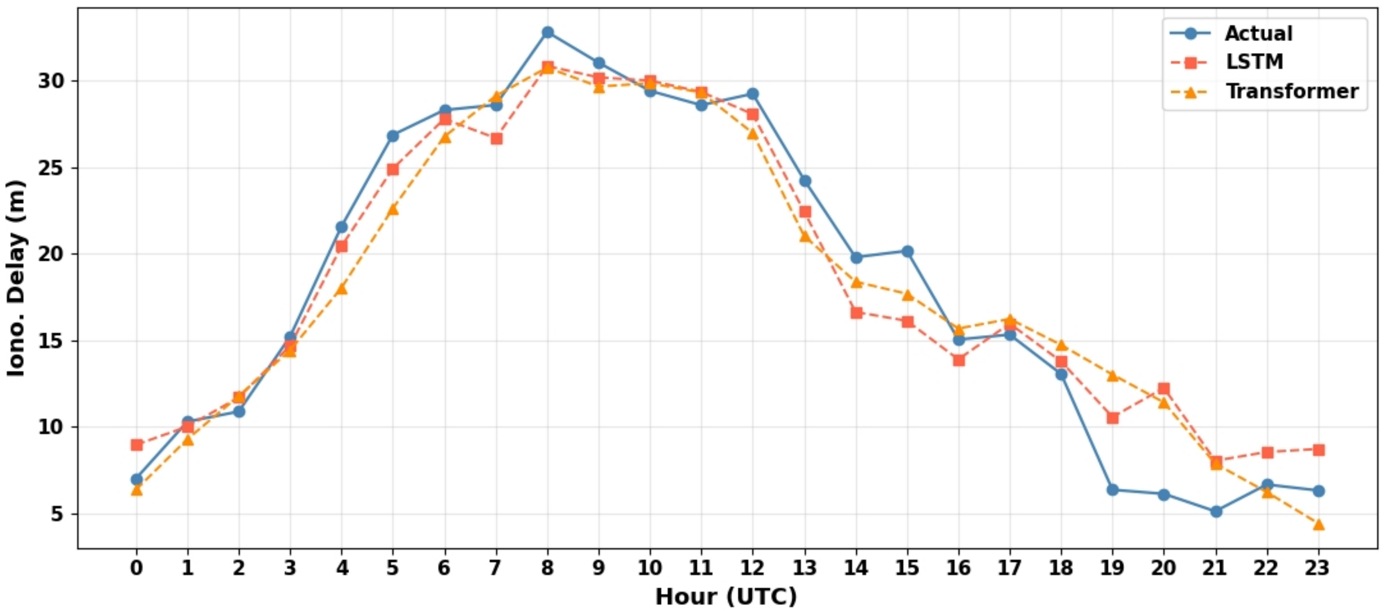
Plot 28- Ionospheric delay (Transformer) 10h May 2024 for L2 Frequency



Plot 29- Ionospheric delay (Transformer) 10h May 2024 for L5 Frequency



Plot 30 - Diurnal hourly profile — 9h May 2024



Plot 31 - Diurnal hourly profile — 10h May 2024